# DIP BHATTACHARJEE

**CSE Undergraduate | AI/ML Researcher | IT Systems & Software**

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github.com/dipjee678

## PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate (CGPA 3.83, Dean's Award) who works on both **AI/ML research** and **practical IT systems**. Author of three IEEE conference papers in 2026 on medical image classification, ML-enabled IoT monitoring, and Monte Carlo risk estimation. Hands-on experience with networking and with installing, configuring and deploying software, from a live multilingual AI CRM on serverless cloud infrastructure to sensor hardware linked to the cloud. Comfortable troubleshooting technical problems and explaining them clearly to non-technical users, from experience as an instructor and tutor. Seeking a role that combines analytical thinking with dependable, secure systems.

## EDUCATION

**Bachelor of Science in Computer Science and Engineering** **Aug 2023 – Present**
*American International University-Bangladesh* *CGPA: 3.83/4.00 · Dean's Award*

**Major:** Computational Theory · **Relevant Courses:** Machine Learning, Natural Language Processing, Graph Theory, Data Structures and Algorithms, Object-Oriented Programming with Java, Programming in C++, Discrete Mathematics.

**HSC, Science** — *Mohipal Government College* *GPA 4.89/5.00 · 2020 – 2022*
**SSC, Science** — *Batisa Secondary School* *GPA 4.89/5.00 · 2018 – 2020*

## RESEARCH & PUBLICATIONS

**Development of a Cloud-Linked Low-Cost Machine Learning-Enabled Weather Monitoring System**
*Published in the Proceedings of IEEE TENSYMP 2026 · DOI: 10.1109/TENSYMP68909.2026.11704037*
Designed a low-cost sensing system that sends readings to the cloud and uses ML to monitor and predict weather conditions, combining hardware, connectivity and data analysis.

**Progressive Two-Stage Transfer Learning with Dual Imbalance Compensation for Interpretable Enamel Caries Classification**
*Accepted for Presentation at the 5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*
Two-stage fine-tuning with imbalance correction at both data and loss level, plus explainable-AI visualisation so clinicians can see where the model looked.

**Diagnostic-Guided Monte Carlo Risk Estimation Using GARCH, EGARCH, and Bootstrap Models**
*Accepted for Presentation at the 3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)*
Statistical diagnostics select between volatility and bootstrap models before Monte Carlo simulation, giving risk estimates that fit the data.

## TECHNICAL SKILLS

### Research & Data

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|---|---|
| **AI & Machine Learning:** | Machine Learning, Deep Learning, Transfer Learning, Classification, Model Training & Evaluation, Explainable AI |
| **NLP & LLMs:** | Text Preprocessing and Classification, LLM APIs (Google Gemini), Structured Outputs, Prompt Design |
| **Data Analysis:** | Data Preprocessing, Analysis and Reporting, Class Imbalance Handling, Monte Carlo Simulation, Statistical Modelling |

### IT & Systems

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|---|---|
| **Networking:** | TCP/IP, IP Addressing and Subnetting, DHCP, DNS, Routing and Switching Basics, LAN/Wi-Fi Setup and Troubleshooting |
| **Software & Deployment:** | Installing, configuring and deploying applications; serverless deployment (Vercel); Git & GitHub |
| **Systems & IoT:** | Sensor hardware integration, cloud connectivity, databases (CRUD applications) |

**Security Practice:** Server-side handling of API keys and secrets, input validation with schemas, safe fallbacks when services fail
**Programming:** Python, C, C++, Java, C#, JavaScript
**Tools:** Microsoft Office, AutoCAD, Adobe Photoshop

## PROJECTS

**LeadDesk: AI Enquiry CRM for Bangla, Banglish and English** *JavaScript, Gemini API, Vercel*

- Built and deployed a web CRM that turns customer messages (WhatsApp, Facebook, web forms) in Bangla, Banglish or English into structured leads using an LLM with schema-enforced outputs.
- Classifies messages (sales, support, spam, job seeker), drafts replies in the customer's language, and tracks follow-ups with a drag-and-drop pipeline, 48-hour reminders and weekly AI summaries.
- Designed for reliability: API key kept on the server, automatic model fallback and retries, and a keyword-based mode that keeps the app working without AI.
- Live demo: biai-crm-eight.vercel.app · Source: github.com/dipjee678/LeadDesk

**Hospital Management System** *Java, OOP*

- Developed a system for patient records, appointments, billing and staff management, using object-oriented design for structured, maintainable code.

**Poultry Farm Management System** *Java, Database*

- Developed a database-driven application with structured storage and full create, read, update and delete workflows for day-to-day farm operations.

## EXPERIENCE

**Instructor** *NextGen Coaching Center for Admission Preparation*

- Taught core admission-preparation subjects and developed mock tests and quizzes to improve students' speed and accuracy.
- Provided one-on-one support, diagnosing where each student was stuck and explaining difficult topics clearly and patiently.

**Freelance Tutor** *Programming & Problem Solving*

- Mentored students in Python, JavaScript and C++, helping them set up their tools, debug their code and work through data structures, algorithms and practical projects.
- Designed personalised learning plans based on each student's level and goals.

## ACHIEVEMENTS

- **Dean's Award**, American International University-Bangladesh
- **Three IEEE conference papers** accepted in 2026 (TENSYMP, SPICSCON, COMPAS), one already published with DOI

## ADDITIONAL INFORMATION

**Languages:** Bangla (Native), English (Fluent), Hindi (Proficient in Listening, Basic Speaking)
**Research Interests:** Machine Learning, NLP, Computer Vision, Explainable AI, Medical Image Analysis, IoT-based Intelligent Systems
**Availability:** Open to internships and roster-based schedules